Anshia Muhammad Yaqoob

437-557-4355 | anshiayaqoob@gmail.com | https://www.linkedin.com/in/anshiamy/ | https://anshiamy.github.io/

Technical Skills

Mechanical: SolidWorks, AutoCAD, Fusion 360, Onshape, GD&T, 3D Printing, CNC Machining
Electronics: KiCad, PCB Design, Power Distribution, Circuit Analysis, Electrical Wiring, Soldering, Oscilloscope, Multimeter
Embedded: Arduino, Raspberry Pi Pico, PLCs, FPGAs, PID Control, Sensor Integration (IMU, Encoder, Ultrasonic), Motor/Servo Control, I/O Integration
Software: C++, Python, Java, JavaScript, VHDL, MATLAB, Git, GitHub, Linux, HTML/CSS

Experience

Technical Specialist May 2026 – Aug 2026
Phoenix Contact Ltd. – Industrial Automation *Milton, ON*

- Independently designed and built 23 customer-specific DIN-rail assemblies from requirements using CLIPX ENGINEER and AutoCAD, defining terminal architecture, power distribution, circuit protection, and bills of materials while integrating and testing 12 automation systems with HMIs, remote I/O, and field devices
- Programmed 11 PLCnext Control systems in PLCnext Engineer using IEC 61131-3 Structured Text and Ladder Logic, implementing control sequences, digital/analog I/O mapping, interlocks, and HMI for customer automation applications
- Configured and validated communications across 5 industrial control networks using Modbus TCP, PROFINET, and OPC UA over Ethernet, performing IP addressing, process-data mapping, and connectivity diagnostics across field devices
- Managed 20–50 daily technical cases through Salesforce and developed product-improvement proposals presented in 5 reviews to product managers and the VP of Automation

Electrical & Systems Integration Lead Sep 2024 – Jul 2025
FIRST Tech Challenge Team 26256 *Mississauga, ON*

- Led a 4-member electrical and systems subteam in developing PID controllers in Java using the FTC SDK for a competition robot's drivetrain and arm subsystems, reducing mean autonomous path-tracking error by 20% over 10 tuning iterations
- Designed and fabricated a custom PCB and wired/troubleshot the robot's electrical architecture, integrating power distribution, Arduino-based servo actuation, IMU, and ultrasonic feedback for autonomous positioning across 6 competition routines
- Designed 12 custom sensor mounts, actuator interfaces, and structural components in SolidWorks, and used 8 AutoCAD layouts to resolve interference, fastener-access, and assembly-clearance constraints before integration
- Executed 42 autonomous validation runs and analyzed control parameters, sensor data, and software revisions in Microsoft Excel, identifying recurring failure modes and using test results to guide calibration and debugging

Projects

Autonomous Line-Following Robot | *Python, Raspberry Pi Pico, PID, SolidWorks*

- Independently designed, fabricated, and wired a custom PCB integrating a Raspberry Pi Pico, 2 infrared sensors, 2 DC motors, and a motor driver, then developed Python control software for autonomous line tracking
- Developed and tuned a closed-loop PID steering controller across 8 iterations, converting sensor-derived tracking error into differential motor commands and reducing mean autonomous path-tracking error by 17%
- Designed and fabricated a SolidWorks chassis across 3 iterations, refining sensor placement, motor mounting, component packaging, and wiring clearances for reliable system integration
- Engineered regulated 5V/3.3V power distribution across 4 electrical subsystems, verifying voltage levels, continuity, and wiring integrity with a multimeter

The Shifting Grove | *Python, Pygame, OOP, BFS/DFS, Finite-State Machines*

- Independently developed a modular Pygame application using object-oriented architecture and centralized state management, integrating reusable entities and 3 self-contained mini-games across single-player and local multiplayer modes
- Implemented procedural maze generation using randomized DFS and loop injection, with BFS-based distance and route validation to ensure reachable objectives and multiple valid paths
- Engineered a six-state autonomous fox agent using BFS pathfinding, predictive pursuit, dynamic replanning, and priority-based state transitions for chase, search, objective interception, and recovery

Education

University of Waterloo Waterloo, ON
BASc Honours Mechatronics Engineering *Sep 2025 – Present*

- **Relevant Coursework:** Introduction to Digital Logic, Circuits, Materials Science, Mechanics of Deformable Solids, Experimental Measurement & Statistical Analysis, Ordinary Differential Equations